LEFT EXACTNESS OF THE FILTERED-COLIMIT EXTENSION OVER A COHERENT RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0GNJ: sources/stacks-chow/tags.txt, source lines 20794–20794. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05CV: sources/stacks-chow/tags.txt, source lines 6933–6933. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05CX: sources/stacks-chow/tags.txt, source lines 6935–6935. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05CW: sources/stacks-chow/tags.txt, source lines 6934–6934. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0034: sources/stacks-chow/tags.txt, source lines 121–121. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0DLP: sources/stacks-chow/tags.txt, source lines 17054–17054. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GNK: sources/stacks-chow/tags.txt, source lines 20795–20795. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0035: sources/stacks-chow/tags.txt, source lines 122–122. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00HA: sources/stacks-chow/tags.txt, source lines 636–636. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:functors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let A, $\mathcal{B}$, F be as in Lemma 0GNJ. Assume A is a coherent ring (Algebra, Definition 05CV), $\mathcal{B}$ is additive, has kernels, filtered colimits commute with taking kernels, and F is left exact. Then F' is additive, left exact, and commutes with arbitrary direct sums.*

Proof. Since A is coherent, the category Mod_A^{fp} is abelian with same kernels and cokernels as in Mod_A, see Algebra, Lemmas 05CX and 05CW. Hence all finite limits exist in Mod_A^{fp} and Categories, Definition 0034 applies. Since F is left exact, F commutes with products of pairs, which are represented by direct sums. Hence F is additive by Homology, Lemma 0DLP. Hence F' is additive and commutes with direct sums by Lemma 0GNK. We urge the reader to prove that F' is left exact themselves instead of reading the proof below.

To show that F' is left exact, it suffices to show that F' commutes with equalizers, see Categories, Lemma 0035. Now, if $a, b : L \to M$ are maps of A-modules, then the equalizer of a and b is the kernel of $a - b : L \to M$. Thus let $0 \to K \to L \to M$ be an exact sequence of A-modules. We have to show that in

$$0 \to F'(K) \to F'(L) \to F'(M)$$

the arrow $F'(K) \to F'(L)$ is a kernel for $F'(L) \to F'(M)$ in $\mathcal{B}$ (if $\mathcal{B}$ were abelian we would say that the displayed sequence is exact). Write $M = \mathrm{colim}_{i \in I} M_i$ as a filtered colimit of finitely presented A-modules, see Algebra, Lemma 00HA. Let $L_i = L \times_M M_i$. We obtain a system of exact sequences $0 \to K \to L_i \to M_i$ over I. Since filtered colimits commute with taking kernels in $\mathcal{B}$ by assumption, it suffices to show that $F'(K) \to F'(L_i)$ is a kernel of $F'(L_i) \to F(M_i)$ in $\mathcal{B}$ for all $i \in I$. In other words, we may assume M is finitely presented. Write $L = \mathrm{colim}_{i \in I} L_i$ as a filtered colimit of finitely presented A-modules. Let $K_i = K \times_L L_i$. We obtain a system of short exact sequences $0 \to K_i \to L_i \to M$ over I. Repeating the argument already given, we reduce to showing $F'(K_i) \to F(L_i)$ is a kernel of $F(L_i) \to F(M)$ in $\mathcal{B}$ for all $i \in I$. In other words, we may assume both L and M are finitely presented A-modules. Since A is coherent, the A-module $K = \mathrm{Ker}(L \to M)$ is of finite presentation as the category of finitely presented A-modules is abelian (see references given above). In other words, all three modules K, L, and M are finitely presented A-modules. This final case follows from the assumption that F is left exact. $\square$